

Combined Drought Indicator (CDI)

The Combined Drought Indicator (CDI) is used to monitor **drought and its stress on vegetation and crop** in Europe. This factsheet provides information about the version 4.1.1 update of the CDI (hereafter, CDIv4.1.1) as implemented in the Copernicus Emergency Management System (CEMS) European Drought Observatory (EDO).

Input Variables	Temporal scale	Spatial scale	Geographic bounding box
Standardized Precipitation Index (SPI)	10 days	1/24 decimal degree (around 5 km at the Equator)	-25 (xmin)
Soil Moisture Index Anomaly (SMA)			51 (xmax)
Fraction of the Photosynthetically Active Radiation (FAPAR) anomaly			22 (ymin)
			72 (ymax)



CDIv4.1 geographic bounding box and study domain (in gray).

1. Brief overview of the indicator

The Combined Drought Indicator is used to identify different classes of drought and their extent with a particular focus on vegetated areas (areas at risk of drought, areas where the vegetation has already been affected and areas where the vegetation is recovering from drought). To this purpose, the CDI integrates three complementary scientific indicators: the **Standardized Precipitation Index (SPI)**, the **Soil Moisture Index Anomaly (SMA)** and the **Fraction of Absorbed Photosynthetically Active Radiation (FAPAR) anomaly**.

The CDI dataset is a collection of raster maps with spatial resolution of around 5km at the Equator and WGS84 (EPSG:4326) as geographic coordinate system. The CDI maps are available from 2012 to present and are regularly updated every 10 days. This is a reasonable frequency for monitoring crop stress, given that usually 10–20 days of water stress could severely affect the final crop production (Rojas, 2021).

Since 2012, the CDI maps and data have been produced operationally and are freely available to final users through the European Drought Observatory web site. In March 2023, **version 3** of the CDI (CDIv3, hereafter) was released. This update featured the incorporation of static crop masks, which prevents the assessment of vegetation stress in crop areas out of the growing season (namely, areas where the crops have not been sown or where they are dormant). The release of the **version 4** (CDIv4, June 2024) takes a further step forward in advancing the performance of the Combined Drought Indicator by featuring:

- the integration of dynamic snow masks for resolving unrealistic negative soil moisture anomalies in presence of snow cover;
- a new Soil Moisture Index Anomaly input dataset based on the European Flood Awareness System (EFAS) version 5.

The CDI **version 4.1** (CDIv4.1, February 2026) improves CDIv4 filtering out extreme FAPAR Anomaly conditions due to excessive moisture.

Finally, the latest CDI **version 4.1.1** (CDIv4.1.1, July 2026) takes into account the transition from SuomiNPP to NOAA-21 in satellite data source for the fAPAR Anomaly component.

2. Agricultural drought

Drought is usually identified with a period characterized by unusually persistent dry weather conditions affecting the hydrological balance. Due to its inherent complexity, different phases/types can be identified and described by using a set of variegated hydro-meteorological and biophysical parameters.

In the academic literature it is common to distinguish at least three phases of drought (Mishra and Singh, 2010):

- **meteorological drought:** typically describing a period with a prolonged precipitation deficit;
- **agricultural drought:** typically describing a period characterized by severe soil moisture anomalies with negative impacts on vegetation and crops;
- **hydrological drought:** when the deficit of water affects rivers, lakes, reservoirs and groundwater.

Agricultural drought is characterized by potential reduced crop health and yield due to meteorological and critical soil conditions. Through the integration of static crop masks, the CDI provides valuable information for assessing the effect of drought in cultivated areas during the growing season.

3. What the indicator shows

The CDI assumes that drought develops as a top-down (hierarchical) process, where a precipitation deficit (**Watch** stage) triggers declining soil moisture conditions (**Warning** stage) which, in turn, promotes negative effects on vegetation, particularly in croplands (**Alert** stage).

From a computational point view, each pixel in a CDI map summarizes: 1) the simultaneous values of three anomaly indicators (SPI, SMA and FAPAR) at the 10-day period **T**; 2) the CDI class assigned to the same pixel in the previous 10-day period **T-1**. Using these two pieces of information, the CDI algorithm assigns each pixel in the study domain to one of the seven classes of Table 1. Each time one or more of the three input indicators (SPI, SMA, FAPAR) fall below (exceed) their corresponding critical thresholds, the CDI classifies the pixel into one of the three primary drought classes (Watch, Warning and Alert). The three **recovery classes** (Recovery, Temporary Soil Moisture Recovery and Temporary Vegetation Recovery) represent temporary stages of transition from drought to "no drought". The purpose of the recovery/temporary classes is twofold. First, they describe "after drought" conditions, namely, pixels where FAPAR anomalies and/or SMA are still negative (after a drought event) but above the reference threshold; second, they allow to partly account for potential asynchronies between spatial anomalies in precipitation, soil moisture and vegetation (Cammalleri et al., 2021).

	LEVEL	INTERPRETATION	LAYER IN EDO MAPVIEWER
0	No drought	Normal conditions (No drought)	No drought and recovery
1	Watch	Precipitation deficit (Issued the $SPI-1 \leq -2$ or the $SPI-3 \leq -1$)	Combined Drought Indicator
2	Warning	Issued when negative soil moisture anomalies are observed, indicated by the $SMA \leq -1$. Warning conditions usually follow precipitation deficit conditions	Combined Drought Indicator
3	Alert	Issued to indicate negative vegetation growth anomalies (FAPAR anomaly ≤ -1), usually in combination with precipitation deficit and negative soil moisture anomalies	Combined Drought Indicator
4	Recovery	Issued immediately after a drought episode, when meteorological conditions, soil water content and vegetation growth return to normal	No drought and recovery
5	Temporary Soil Moisture recovery	Issued after a drought episode: soil moisture conditions are above the critical threshold ($SMA > -1$) but not enough to consider the episode closed	No drought and recovery
6	Temporary vegetation recovery	Issued after a drought episode: vegetation conditions are above the critical threshold (FAPAR anomaly > -1) but not enough to consider the episode closed	No drought and recovery

Table 1. In the CDI classification scheme, each level is identified by a colour in order to facilitate its interpretation and communication. It is worthwhile to observe that the increasing drought severity from Watch to Alert is represented through warm colors (yellow, orange, red), a common convention for signaling danger levels (Verpe Engeset et al., 2022).

Note for users of the on-line map viewer: The default display of the CDI in the EDO map viewer shows the "Combined Drought Indicator" layer (right column in Table 1). This is to make the interpretation of the maps easier for users. However, advanced users have the option to display also the "No drought and recovery" layer according to their preferences.

4. How the indicator is calculated

The CDI combines anomalies of precipitation, soil moisture and vegetation greenness (see the CDI conceptual framework, Table 2a). More specifically, the CDI uses three drought indicators and two thematic masks implemented in EDO:

- ❑ **Standardized Precipitation Index (SPI):** this is a well-known indicator recommended by the World Meteorological Organization (WMO) for meteorological drought monitoring. Generally speaking, the SPI quantifies the deviation of the cumulated precipitation on a given timescale with respect to a **reference** baseline (McKee et al., 1993; Edwards and McKee, 1997). The CDI computation considers the SPI at two different timescales: a 30-day (SPI-1) and a 90-day timescale (SPI-3). Below-normal conditions in the precipitation level are assessed when one or both of these two indicators fall below their corresponding critical thresholds: a value of -1 for the SPI-3 ("*Moderate meteorological drought*") and -2 for the SPI-1 ("*Extreme meteorological drought*"). The SPI-1 and SPI-3 indicators are calculated using precipitation data from the ERA5 (European ReAnalysis 5th Generation) reanalysis (Hersbach et al., 2017).
- ❑ **Soil Moisture Index Anomaly:** this indicator is derived from anomalies of estimated daily soil moisture (or soil water) content, represented as standardized Soil Moisture Index (SMI). The SMI is a product from the JRC LISFLOOD hydrological model (de Roo et al. 2000; Laguardia and Niemeyer, 2008). The current SMI product is based on the European Flood Awareness System (EFAS) version 5.0, operationally released on the EFAS-IS on 20 September 2023.
- ❑ **FAPAR anomaly:** FAPAR represents the fraction of incident solar radiation that is absorbed by land vegetation for photosynthesis. Satellite-derived FAPAR anomalies can be effectively used for detecting and assessing the impact of drought on different land cover types (Gobron et al., 2005).
- ❑ **Crop Masks:** In regions where agricultural production has strong seasonal patterns, the soil surface experiences shifts from vegetation to bare soil (after crop harvest) which significantly affect FAPAR anomalies in cropland areas. This means that, if not properly masked, anomalous signals of FAPAR from exposed bare soil in crop fields risk to be confused with drought stress on vegetation, resulting in wrong "Alert" assessments. The crop masks used for computing the CDI consist in a collection of 10-day gridded binary maps, where each pixel indicates the presence or absence of crop during the growing season. The crop masks allow to limit the analysis of negative FAPAR anomalies in crop fields to the growing season (see the CDI conceptual framework, Table 2b)
- ❑ **Snow Masks:** The function of the snow mask in assessing soil moisture anomalies is comparable to that of the crop masks in evaluating FAPAR anomalies. Under snow cover, soil moisture anomalies could be unreliable. If not properly masked, anomalous signals of soil moisture from snow-covered ground risk to trigger wrong "Warning" assessments. To overcome this limitation, the CDIv4 integrates a dynamic snow cover masking process. This consists in applying a gridded binary map, where each pixel indicates the presence or absence of snow on the ground for each 10-day period. When pixels with snow are detected, the corresponding soil moisture and FAPAR anomalies are excluded from the CDI (see the CDI conceptual framework, Table 2c).

5. CDI conceptual framework

In a CDI map each pixel must be intended as the result of an algorithm which implements the logic in Table 2a. This displays all the possible combinations among inputs at time step **T-1** and **T**, as well as their related outputs. The CDI conceptual framework is divided into eight-lettered columns (A through H) and seven-numbered rows. Each column details how the three input anomaly indicators SPI, SMA and FAPAR Anomaly intersect at time step **T**; the rows indicate the drought conditions (CDI levels) at time step **T-1**; the cells (the combination among columns and rows) return the CDI level at time step **T**. In each column the variables are arranged in a hierarchical structure which reflects the possible evolution process from meteorological drought to agricultural drought (SPI $\Rightarrow$ SMA $\Rightarrow$ FAPAR Anomaly). In addition, for each indicator the reference thresholds are reported in the column headings.

The conceptual framework introduced in Table 2a outlines the calculation process of the CDI when all the three input anomaly indicators (SPI, SMA, FAPAR anomaly) are considered. However, the use of snow masks implies the exclusion of SMA where the pixels are masked (pixels with snow on the ground). Furthermore, if we also (reasonably) assume that in those pixels negative vegetation anomalies cannot occur, then the CDI conceptual framework boils down to the two-column Table 2c, where both negative SMA and FAPAR anomalies are excluded. This means that, in presence of snow, the soil moisture and the vegetation conditions are considered unreliable and excluded from the computation process. Similarly to what explained for snow masks, the use of crop masks implies that negative FAPAR anomalies are excluded. The corresponding conceptual framework is illustrated in Table 2b.

We conclude this section by noting that variations in vegetation health and/or cover may also be driven by stress factors not directly related to drought conditions (e.g., plant diseases, pests, hail, or flooding). With specific regard to flooding, although water is essential for plant growth, excessive water, waterlogging, or flooding can negatively affect plant development throughout the lifecycle and ultimately reduce yields (Zhou et al., 2020). To account for this, CDIv4.1 excludes negative FAPAR anomalies when SPI-1 is greater than 1, as such conditions indicate excessive moisture rather than drought-related stress.

			Drought Indicators at the 10-day period T																		
			A			B			C				D	E	F			G	H		
			=0			=1			=0		!(SPI-1>0.5)		=0		=1						
			(SPI-1>0.5)			(SPI-1>0.5)			AND		AND										
			(SPI-3>0)			(SPI-3>0)			(SPI-3>0)		(SPI-3>0)										
			>-1			>-1			>-1				>-1	>-1	>-1			>-1	>-1		
			≤ -0.5	(-0.5;0]	>0	≤ -0.5	(-0.5;0]	>0	≤ -1				>-1	≤ -1	≤ -1			>-1	≤ -1		
			>-1			>-1			>-1				≤ -1	≤ -1	>-1			≤ -1	≤ -1		
			≤ -0.5	(-0.5;0]	>0	≤ -0.5	(-0.5;0]	>0	≤ -0.5	(-0.5;0]	(-0.5;0]	>0	≤ -1	≤ -1	≤ -0.5	(-0.5;0]	>0	≤ -1	≤ -1		
			No			Watch			No		Warning		No	Alert							
			Drought			Drought			Drought		Warning		Drought	Alert							
			Recovery			Watch															
			Warning		Temp. SM Recovery	Recovery	Warning		Temp. SM Recovery	Watch	Warning						Warning				
			Recovery				Watch				Warning						Warning				
			Alert		Temp. Vegetation Recovery		Recovery	Alert			Temp. Vegetation Recovery	Warning	Alert		Warning	Alert		Temp. Vegetation Recovery	Warning	Alert	
			Recovery					Watch			Warning					Warning		Warning			

Table 2a. CDI conceptual framework. The colour scheme is the same as used in Table 1. Note, to improve the readability of the table, we use "0" and "1" as concise representations of "(SPI-1 > -2) AND (SPI-3 > -1)" and "(SPI-1 < -2) OR (SPI-3 < -1)" respectively.

CDI level in the previous 10-day period T-1			Drought Indicators at the 10-day period T												
			A			B			C		F				
			SPI	=0			=1			=0		=1			
										(SPI-1 >0.5)	!(SPI-1 >0.5)				
										AND	AND				
			SMA	>-1			>-1			≤ −1		≤ −1			
													≤ −0.5	(-0.5;0]	>0
													≤ −0.5	(-0.5;0]	>0
			1	No Drought	No			Watch			No	Warning			
2	Recovery	Drought						Drought							
3	Watch	Recovery			Watch										
4	Warning	Warning	Temp. SM Recovery	Recovery	Warning	Temp. SM Recovery	Watch			Warning					
5	Temp. SM Recovery	Recovery			Watch			Watch							
6	Alert							Warning							
7	Temp. Vegetation Recovery	Recovery			Watch										

Table 2b. CDI conceptual framework for pixels where crop masks are applied. When crop fields are out of the growing season, negative FAPAR anomalies are not involved in computing the Combined Drought Indicator.

			Drought Indicators at the 10-day period T	
			A	B
			=0	=1
CDI level in the previous 10-day period T-1	1	No Drought	No	Watch
	2	Recovery	Drought	
	3	Watch		
	4	Warning		
	5	Temp. SM Recovery		
	6	Alert	Recovery	
	7	Temp. Vegetation Recovery		

Table 2c. CDI conceptual framework for pixels where snow masks are applied. When snow is observed on the ground, negative SMA and FAPAR anomalies are not involved in computing the Combined Drought Indicator.

5. How to use the indicator

In this section, we show with an example how the simultaneous application of crop and snow masks helps to reduce unreliable spatial patterns in the CDI maps. We consider the spatio-temporal evolution of the CDI in three Nordic countries (Finland, Sweden and Norway), characterized by a relatively small crop production. The period of interest is the first six months of year 2020, when the CDlv3 (Figure 1) exhibits severe dry conditions, with several pixels assigned to the "Warning" class (soil moisture deficit) gradually evolving to the "Alert" class (vegetation stress). The eighteen 10-day maps in Figure 2 display the spatial distribution of the crop fields (active and non-active crop fields) in the January-June period. A visual analysis of these maps highlights that most of the pixels in Alert condition since early May 2020 are associated to negative FAPAR anomalies in active crop fields. It is worthwhile to observe that part of these pixels (especially in the northern part of Sweden and Finland) also depend on negative FAPAR anomalies from vegetation other than crop. As soil moisture anomalies based on the LISFLOOD hydrological model are not reliable in snow covered areas (Figure 3), drought signals in Figure 1 can convey misleading messages to final users. The CDlv4 overcomes this issue masking out the snow-covered area and simplifying the CDI conceptual framework from Table 2a to Table 2c. The maps in Figure 4 show that, where the snow masks are applied, the Warning and Alert signals disappear. However, when negative soil moisture anomalies appear in snow ground (e.g. in June 2020), the spatial patterns of the CDlv3 and CDlv4 maps are almost identical.

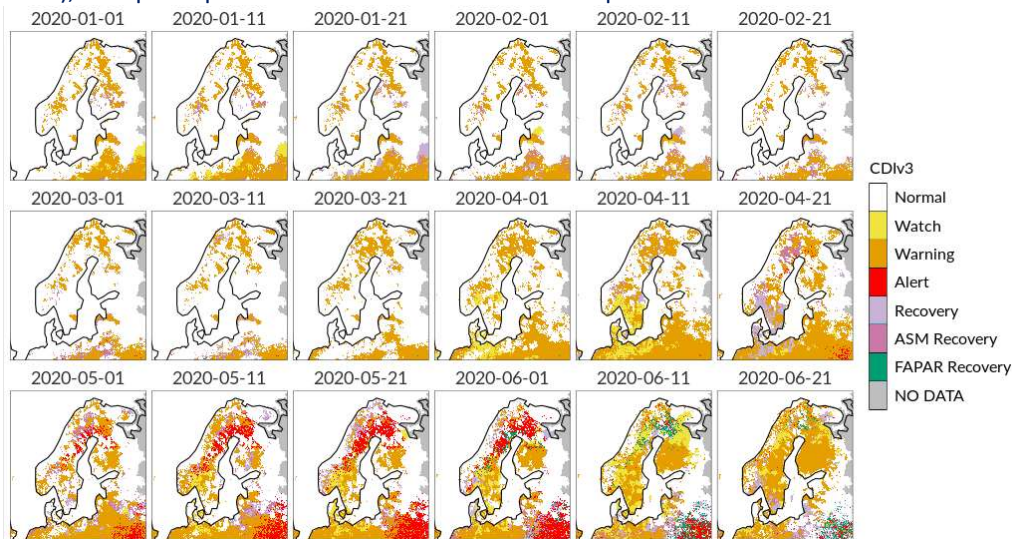


Figure 1. Combined Drought Indicator version 3 for Finland, Sweden and Norway during the 1st to the 18th 10-day period in 2020 (January - June).

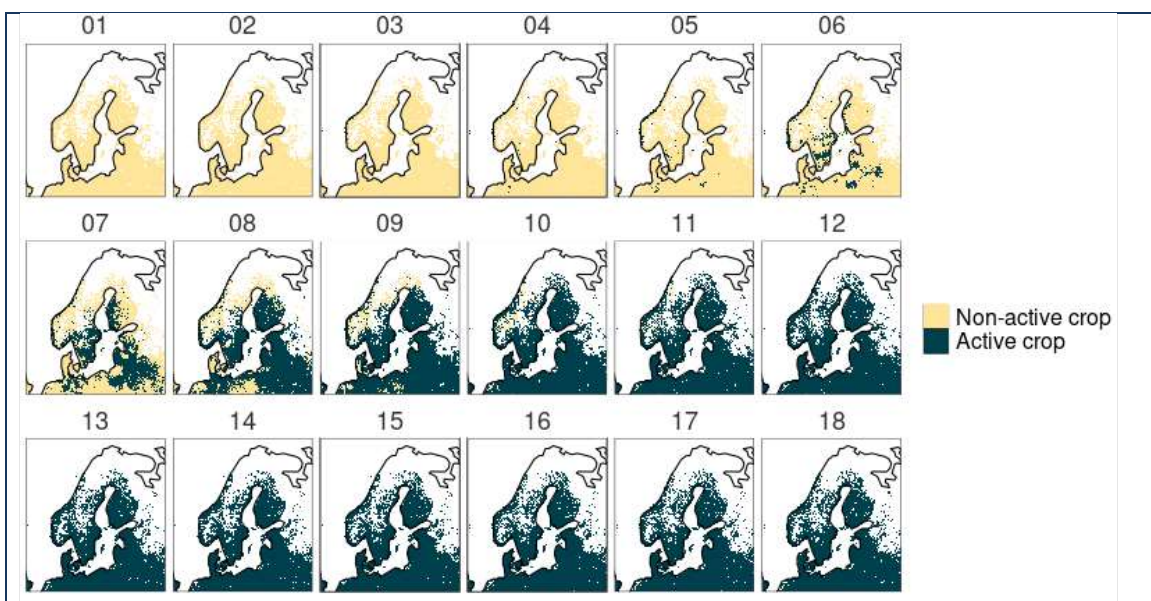


Figure 2. Static crop masks for Finland, Sweden and Norway during the 1st to the 18th 10-day period in a year (January – June).

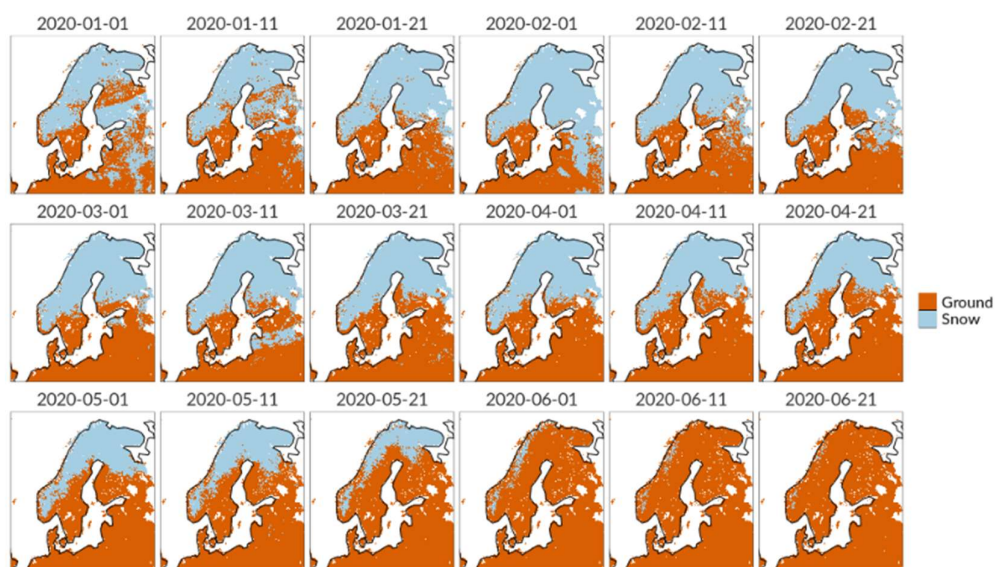


Figure 3. Dynamic snow masks for Finland, Sweden and Norway during the 1st to the 18th 10-day period in 2020 (January – June).

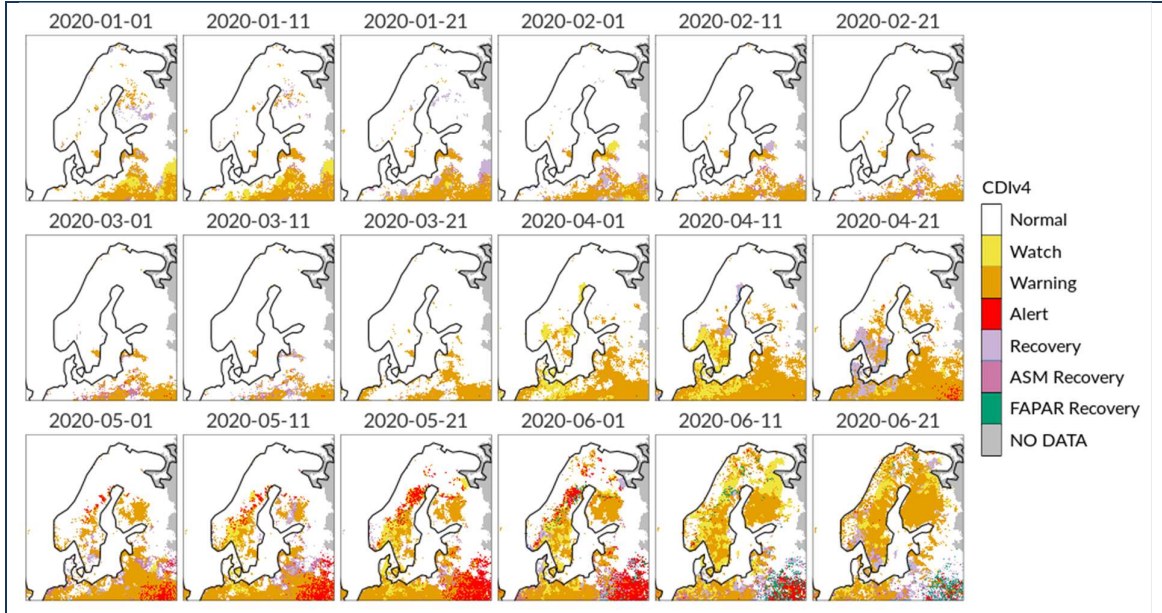


Figure 4.1 Combined Drought Indicator version 4 for Finland, Sweden and Norway during the 1st to the 18th 10-day period in 2020 (January - June).

6. Strengths and weaknesses of the indicator

Strengths:

- By integrating meteorological, hydrological and vegetation data, a combined indicator can help reduce false alarms in drought assessment. For example, a biomass reduction can be caused by factors other than drought-induced water stress.
- Evidence of drought based on an integrated approach can support policy-makers in effective risk management and decision-making.
- The use of spatial indicators of phenology (crop masks) provides more robust assessment of potential drought impacts on crops.

Weaknesses:

- The satellite-derived FAPAR Anomaly indicator is based on reflected solar radiation, with wavelengths in the optical region (i.e., visible and infrared wavelength) of the electromagnetic spectrum, and is therefore not effective in the presence of clouds. Clouds are generally masked out before the indicator is computed. However, low clouds are not always detected, resulting in erroneous indicator values. This problem is particularly in northern Europe. One way to address this issue would be to use more than one indicator related to vegetation growth.
- The CDI signal over snow covered regions comes from a real precipitation deficit combined with a soil moisture deficit estimated by hydrological simulations. Modelled soil moisture has a low reliability in snow cover areas especially where the snow season is long. The Hydrological model currently in use needs to be improved considering the snow dynamics. In this respect, future developments of the model are in progress. In the short term, the CDIV4 integrates snow cover data.
- While the snow masks are dynamic maps, the CDI crop masks provide "static" and "generic" land cover information, given that the start and the end of the growing season of each crop cell are fixed in time (don't change from one year to the other).

References

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Indicators

- **Standardized Precipitation Index (SPI):**
https://drought.emergency.copernicus.eu/data/factsheets/factsheet_spi.pdf
<https://data.jrc.ec.europa.eu/dataset/ca9f628f-8595-4539-863a-fc9285499496>
<https://data.jrc.ec.europa.eu/dataset/29fdaa3d-c7ba-4a47-a130-dda7c26a0fe1>
- **Soil Moisture Anomaly:**
https://drought.emergency.copernicus.eu/data/factsheets/factsheet_soilmoisture.pdf
<https://data.jrc.ec.europa.eu/dataset/882501f9-b783-4b6e-8aca-1875a7c0b372>
- **fAPAR Anomaly:**
https://drought.emergency.copernicus.eu/data/factsheets/factsheet_fapar_viirs.pdf
<https://data.jrc.ec.europa.eu/dataset/ea144baf-f8f4-4451-aeefc-fc20edcc8cc7>
- **Crop masks:**
https://drought.emergency.copernicus.eu/data/factsheets/factsheet_crop_masks.pdf
<https://data.jrc.ec.europa.eu/dataset/d0fe3beb-1478-44bf-b384-83f8780e2685>

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